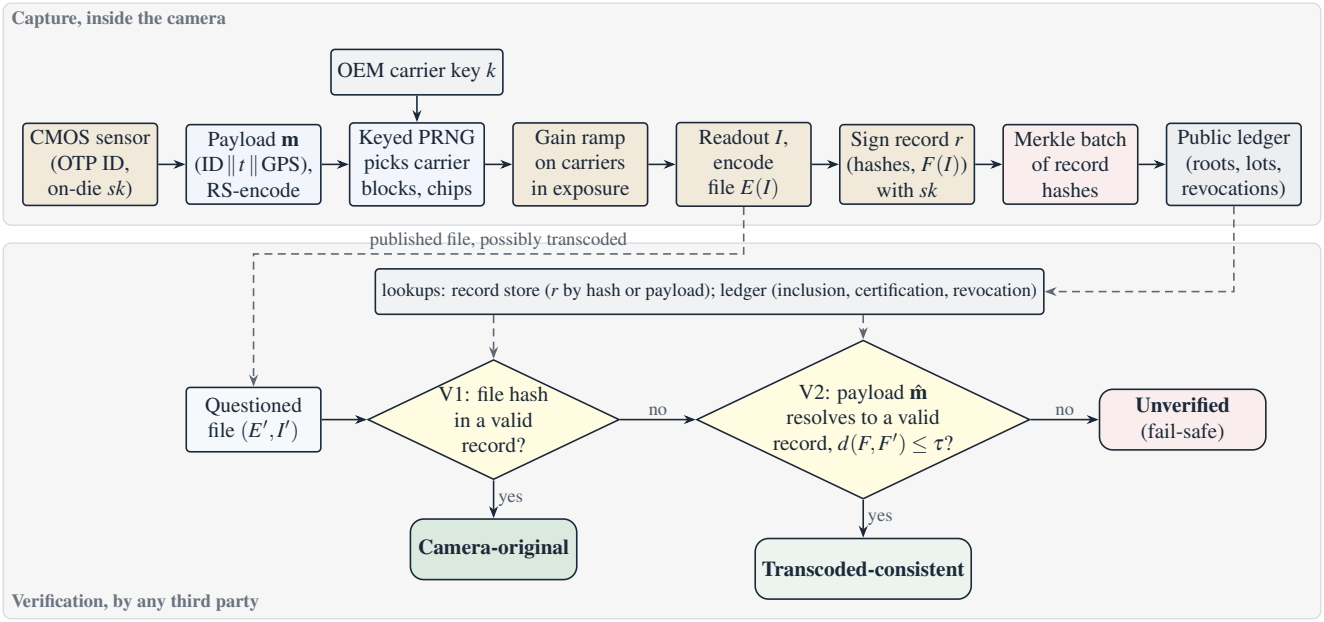

Capture, inside the camera
OEM carrier key k
CMOS sensor (OTP ID, on-die sk)
Payload $\mathbf{m}$ (ID $\|$ t $\|$ GPS), RS-encode
Keyed PRNG picks carrier blocks, chips
Gain ramp on carriers in exposure
Readout I, encode file $E(I)$
Sign record r (hashes, $F(I)$) with sk
Merkle batch of record hashes
Public ledger (roots, lots, revocations)
published file, possibly transcoded
lookups: record store (r by hash or payload); ledger (inclusion, certification, revocation)
Questioned file (E', I')
V1: file hash in a valid record?
no
V2: payload $\hat{\mathbf{m}}$ resolves to a valid record, $d(F, F') \le \tau$?
no
Unverified (fail-safe)
yes
Camera-original
yes
Transcoded-consistent
Verification, by any third party